

WEEK 3 PRACTICAL

Task 1: Document Modelling

Documents 4

Aggregations

Schema

Indexes 1

🕒 ▼

Type a query: { field: 'value' } or [Generate query](#)

+ ▼

📄 ▼

🔗 ▼

</>

▶

```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
  ▶ formats: Array (3)
  ▶ specs: Object (5)
    avgRating: 4.6
  ▶ reviews: Array (2)
```

```
_id: ObjectId('6a8d09b39109beb4f7521e63')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
  ▶ formats: Array (2)
  ▶ specs: Object (3)
    avgRating: 4.2
  ▶ reviews: Array (2)
```

```
_id: ObjectId('6a8d09c09109beb4f7521e65')
title: "Whispers of Yesterday"
author: "S. Kapoor"
genre: "Fiction"
publishedYear: 2019
▸ formats: Array (2)
▸ specs: Object (2)
  avgRating: 3.8
▸ reviews: Array (2)
```

```
_id: ObjectId('6a8d09cb9109beb4f7521e67')
title: "Quantum Horizons"
author: "R. Menon"
genre: "Science"
publishedYear: 2023
▸ formats: Array (2)
▸ specs: Object (4)
  avgRating: 4.9
▸ reviews: Array (2)
```

```
> use("BookFlow")
< switched to db BookFlow
> db["book_metadata"].find()
< {
  _id: ObjectId('6a8d09a79109beb4f7521e61'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
}
```

```
    fileSize_MB: 5.6,  
    drm: true,  
    narrator: 'P. Iyer',  
    runtime_min: 540  
  },  
  avgRating: 4.9,  
  reviews: [  
    {  
      member_id: 'M4001',  
      rating: 5,  
      comment: 'The Digital format made annotations so easy.',  
      date: 2023-08-01T00:00:00.000Z  
    },  
    {  
      member_id: 'M4002',  
      rating: 5,  
      comment: 'Mind-bending and well narrated.'
```

Task 2: Advanced Querying

🕒 { "avgRating": { "\$gt": 4 } }

[Generat](#)



▶ `_id: ObjectId('6a8d09a79109beb4f7521e61')`
`title: "The Silent Algorithm"`
`author: "R. Menon"`
`genre: "Technology"`
`publishedYear: 2022`
▶ `formats: Array (3)`
▶ `specs: Object (5)`
`avgRating: 4.6`
▶ `reviews: Array (2)`

`_id: ObjectId('6a8d09b39109beb4f7521e63')`
`title: "Ocean of Code"`
`author: "A. Fernandes"`
`genre: "Technology"`
`publishedYear: 2021`
▶ `formats: Array (2)`
▶ `specs: Object (3)`
`avgRating: 4.2`
▶ `reviews: Array (2)`

`_id: ObjectId('6a8d09cb9109beb4f7521e67')`
`title: "Quantum Horizons"`
`author: "R. Menon"`
`genre: "Science"`
`publishedYear: 2023`
▶ `formats: Array (2)`
▶ `specs: Object (4)`
`avgRating: 4.9`
▶ `reviews: Array (2)`

Generat



```
  _id: ObjectId('6a8d09a79109beb4f7521e61')
  title: "The Silent Algorithm"
  author: "R. Menon"
  genre: "Technology"
  publishedYear: 2022
  formats: Array (3)
  specs: Object (5)
    avgRating: 4.6
  reviews: Array (2)
```



```
  _id: ObjectId('6a8d09b39109beb4f7521e63')
  title: "Ocean of Code"
  author: "A. Fernandes"
  genre: "Technology"
  publishedYear: 2021
  ▶ formats: Array (2)
  ▶ specs: Object (3)
    avgRating: 4.2
  ▶ reviews: Array (2)
```

```
_id: ObjectId('6a8d09cb9109beb4f7521e67')
title: "Quantum Horizons"
author: "R. Menon"
genre: "Science"
```

Generate

25

```

  ▶ _id : ObjectId('6a8d09a79109beb4f7521e61')
    title : "The Silent Algorithm"
    author : "R. Menon"
    genre : "Technology"
    publishedYear : 2022
  ▶ formats : Array (3)
  ▶ specs : Object (5)
    avgRating : 4.6
  ▶ reviews : Array (2)

```

```
{
  "avgRating": {
    "$gt": 4
  },
  "formats": {
    "$in": [
      "Digital"
    ]
  }
}
```



```
▶ reviews : Array (2)
```

```
_id: ObjectId('6a8d09cb9109beb4f7521e67')
title: "Quantum Horizons"
author: "R. Menon"
genre: "Science"
publishedYear: 2023
▶ formats: Array (2)
▶ specs: Object (4)
  avgRating: 4.9
▶ reviews: Array (2)
```

🕒 ▼

```
{
  "formats": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

🔗



▶

```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
▸ formats: Array (3)
▸ specs: Object (5)
  avgRating: 4.6
▸ reviews: Array (2)
```

🕒 ▼

```
{
  "reviews.comment": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```



▶

```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
▸ formats: Array (3)
▸ specs: Object (5)
  avgRating: 4.6
▸ reviews: Array (2)
```

Task 3: Aggregation Pipeline — "Top Rated" Report

Documents 5

Aggregations

Schema

Indexes 1

Validation

▼

\$match

Generate aggregation ⚡💡

Explain

Run

Untitled - modified

SAVE

+ CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

⌵

Stage 1 \$match

1 {

2 "publishedYear": {

3 "\$gt": 2020

4 }

5 }

Output preview after \$match stage (Sample of 3 documents)

▶

_id: ObjectId('6a8d09a79109beb4f7521e61')

title: "The Silent Algorithm"

author: "R. Menon"

genre: "Technology"

publishedYear: 2022

formats: Array (3)

specs: Object (5)

avgRating: 4.6

reviews: Array (2)

Documents 5

Aggregations

Schema

Indexes 1

Validation

▼

\$match

Generate aggregation ⚡💡

Explain

Run

Options ▶

Untitled - modified

SAVE

+ CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

⌵

Stage 1 \$match

1 {

2 "publishedYear": {

3 "\$gt": 2020

4 }

5 }

Output preview after \$match stage (Sample of 3 documents)

j('6a8d09b39109beb4f7521e63')

n of Code

n of Code

Fernandes

nology"

r: 2021

ay (2)

t (3)

.2

av (2)

▶

_id: ObjectId('6a8d09cb

title: "Quantum Horizor

author: "R. Menon"

genre: "Science"

publishedYear: 2023

formats: Array (2)

specs: Object (4)

avgRating: 4.9

reviews: Array (2)

▼

\$match

\$unwind

Generate aggregation ⚙️💡

Explain

Run

Untitled - modified

SAVE

+ CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

▼ Stage 2 | \$unwind

1 "\$reviews"

Output preview after \$unwind stage (Sample of 6 documents)

▶

```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Object (4)
```

Documents 5

Aggregations

Schema

Indexes 1

Validation

▼

\$match

\$unwind

Generate aggregation ⚙️💡

Explain

Run

Untitled - modified

SAVE

+ CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

▼ Stage 2 | \$unwind

1 "\$reviews"

Output preview after \$unwind stage (Sample of 6 documents)

▶

```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Object (4)
```

 \$match \$unwind \$groupGenerate aggregation  

Explain

Run

Untitled - modified

 SAVE

+ CREATE NEW

 EXPORT DATA

</> EXPORT CODE

 PREVIEW

{ } STAGES

Stage 3 \$group



```
1 {
2   "_id": "$title",
3   "author": {
4     "$first": "$author"
5   },
6   "avgReviewRating": {
7     "$avg": "$reviews.rating"
8   },
9   "totalReviews": {
10    "$sum": 1
11  }
12 }
```



Output preview after \$group stage (Sample of 3 documents)

```
_id: "Quantum Horizons"
author : "R. Menon"
avgReviewRating : 5
totalReviews : 2
```

 \$match \$unwind \$groupGenerate aggregation  

Explain

Run

Untitled - modified

 SAVE

+ CREATE NEW

 EXPORT DATA

</> EXPORT CODE

 PREVIEW

{ } STAGES

Stage 3 \$group



```
1 {
2   "_id": "$title",
3   "author": {
4     "$first": "$author"
5   },
6   "avgReviewRating": {
7     "$avg": "$reviews.rating"
8   },
9   "totalReviews": {
10    "$sum": 1
11  }
12 }
```



Output preview after \$group stage (Sample of 3 documents)

```
_id: "The Silent Algorithm"
author : "R. Menon"
avgReviewRating : 4.5
totalReviews : 2
```

Stage 4 \$sort



```
1 {
2   "avgReviewRating": -1
3 }
```



Output preview after \$sort stage (Sample of 3 documents)

```
_id: "Quantum Horizons"
author : "R. Menon"
avgReviewRating : 5
totalReviews : 2
```

Untitled - modified SAVE + CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES TEXT

```
1 {
2   "avgReviewRating": -1
3 }
```

Output preview after [\\$sort](#) stage (Sample of 3 documents)

```
_id: "Ocean of Code"
author: "A. Fernandes"
avgReviewRating: 4.5
totalReviews: 2
```

```
_id: "The S
author: "R
avgReviewRa
totalReview
```

\$match \$unwind \$group \$sort \$project Generate aggregation Explain Run

Untitled - modified SAVE + CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES

Stage 5 [\\$project](#)

```
1 {
2   "_id": 0,
3   "title": "$_id",
4   "author": 1,
5   "avgReviewRating": {
6     "$round": [
7       "$avgReviewRating",
8       2
9     ]
10  },
11  "totalReviews": 1
12 }
```

Output preview after [\\$project](#) stage (Sample of 3 documents)

```
author: "R. Menon"
totalReviews: 2
title: "Quantum Horizons"
avgReviewRating: 5
```

\$match \$unwind \$group \$sort \$project Generate aggregation Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES TEXT

Stage 5 [\\$project](#)

```
1 {
2   "_id": 0,
3   "title": "$_id",
4   "author": 1,
5   "avgReviewRating": {
6     "$round": [
7       "$avgReviewRating",
8       2
9     ]
10  },
11  "totalReviews": 1
12 }
```

Output preview after [\\$project](#) stage (Sample of 3 documents)

```
: "A. Fernandes"
views: 2
"Ocean of Code"
iewRating: 4.5
```

```
author: "R. Menor
totalReviews: 2
title: "The Siler
avgReviewRating:
```

Task 4: Indexing for Speed

Documents 5

Aggregations

Schema

Indexes 2

Validation

Create Index

Refresh

VIEWING

INDEXES

SI

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	9 (since Tue Aug 25 2026)	UNIQUE	READY
> reviews.comment_text	TEXT	20.5 kB	0 (since Tue Aug 25 2026)		READY

Documents 5

Aggregations

Schema

Indexes 2

Validation

🕒

{
 "\$text": {
 "\$search": "Digital"
 }
}

Generate query ⚡

Explain

+ ▾

📄 ▾

🔗 ▾

</>

25 ▾

1 - 3 of 3

↺

```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```



```
{
  "$text": {
    "$search": "Digital"
  }
}
```

[Generate query](#)[Explain](#)[Reset](#)

25

1 - 3 of 3



```
genre : "Technology"
publishedYear : 2023
formats : Array (2)
specs : Object (4)
avgRating : 4.9
reviews : Array (2)
```

```
_id: ObjectId('6a8d09b39109beb4f7521e63')
title : "Ocean of Code"
author : "A. Fernandes"
genre : "Technology"
publishedYear : 2021
formats : Array (2)
specs : Object (3)
avgRating : 4.2
reviews : Array (2)
```



```
{
  "reviews.comment": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

[Generate query](#)

25

1 - 3



```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title : "The Silent Algorithm"
author : "R. Menon"
genre : "Technology"
publishedYear : 2022
formats : Array (3)
specs : Object (5)
avgRating : 4.6
reviews : Array (2)
```

```
_id: ObjectId('6a8d09b39109beb4f7521e63')
title : "Ocean of Code"
author : "A. Fernandes"
genre : "Technology"
publishedYear : 2021
```

Create Index

Refresh

VIEWING

INDEXES

SI

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR <i>i</i>	36.9 kB	9 (since Tue Aug 25 2026)	UNIQUE <i>i</i>	READY
> reviews.comment_text	TEXT <i>i</i>	20.5 kB	2 (since Tue Aug 25 2026)		READY
> Field: formats_1	REGULAR <i>i</i>	20.5 kB	0 (since Tue Aug 25 2026)		READY

Create Index

Refresh

VIEWING

INDEXES

SEARCH INDEXES

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR <i>i</i>	36.9 kB	9 (since Tue Aug 25 2026)	UNIQUE <i>i</i>	READY
> reviews.comment_text	TEXT <i>i</i>	20.5 kB	2 (since Tue Aug 25 2026)		READY
> Field: formats_1	REGULAR <i>i</i>	20.5 kB	0 (since Tue Aug 25 2026)		READY
> Field: avgRating_1	REGULAR <i>i</i>	20.5 kB	0 (since Tue Aug 25 2026)		READY



```
{
  "avgRating": {
    "$gt": 4
  }
}
```

[Generate query](#)

Expla



25 ▾

1 - 3 of 3



```
_id: ObjectId('6a8d09a79109beb4f7521e61')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
▶ formats: Array (3)
▶ specs: Object (5)
  avgRating: 4.6
▶ reviews: Array (2)
```

```
_id: ObjectId('6a8d09b39109beb4f7521e63')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
▶ formats: Array (2)
```